# AFLDiff: Adaptive Frequency-Aware Latent Diffusion Model for Road Extraction in Remote Sensing Images

Ruiqi Liu, Qusen Chen, Yang Liu, Zidi Shi, Zhao Li, Weiping Jiang, and Dawei Li

***Abstract*—Accurate road extraction from remote sensing images is of great importance, yet the complex backgrounds, significant scale differences, and detailed characteristics of road information pose a tremendous challenge to the semantic modeling capabilities of deep learning methods. Recent studies have shown impressive success with diffusion-based methods in semantic segmentation. However, when applied to road extraction tasks, the remarkable performance can be diminished by intricate terrains and unreasonable feature fusion. In this article, we reframe road extraction as a mask-generating task, progressively refining road mask predictions through a timestep-by-timestep denoising process. Specifically, we propose a novel adaptive frequency-aware latent diffusion framework named AFLDiff. The pretrained VQ-VAE encoder with coupled perceptual compression suppresses semantic features that are weakly associated with roads in remote sensing images, reconstructing the image into a low-dimensional latent space for training. The adaptive frequency-aware module filters out redundant noise from the noisy features and enhances useful high- and low-frequency components, bridging the gap between noisy and semantic features for better fusion. Extensive experimental results demonstrate that our method achieves state-of-the-art (SOTA) performance on the DeepGlobe, Massachusetts, and CHN6-CUG datasets, with cross-dataset validation further confirming its robustness and generalization capabilities.**

***Index Terms*—Adaptive frequency-aware, diffusion model, latent space, perceptual compression, road extraction.**

## I. Introduction

ROADS are a key terrestrial feature that plays a vital role in urban planning and traffic management [1]. Remote sensing technology enables rapid and effective collection of geographic information through distant observation of the Earth's surface over vast areas [2]. Accurately extracting roads from widely covered and regularly updated remote sensing images helps establish well-structured road networks, which offer a solid data foundation for autonomous driving and intelligent navigation [3]. With the development of computer vision, deep learning techniques have increasingly automated the learning of unified feature representations from large-scale remote sensing data, gradually becoming the primary methods for road extraction from remote sensing images [4].

Within high-resolution remote sensing images, the complex background, the significant scale differences, and the intricate branching details in roads pose tremendous challenges to the semantic understanding capabilities of deep learning models. Most conventional road extraction studies fall into two categories: segmentation-based methods and graph-inference-based methods. The existing research has significantly improved benchmarks on public datasets. However, on one hand, the pixel-level discrimination of segmentation models makes it difficult to ensure the topological integrity of the segmentation results [5]. On the other hand, graph inference modules are susceptible to error accumulation due to local inference mistakes, leading to poor overall prediction accuracy [6]. Although our previous work, TPEGAN [7], effectively combines pixel-level enhancement and graph inference to achieve a fine balance between connectivity and accuracy, it still struggles to recognize details due to limitations in semantic perception capabilities. The core issue is that these methods typically aim to learn elaborated image features, with their training mode focusing on maximizing the posterior probability $p(x|y)$ that predicts the mask $x$ given the input image $y$, while in this process, the intrinsic characteristics of $x$ are overlooked [8].

In recent years, denoising diffusion probabilistic models have demonstrated outstanding performance in various conditional and unconditional image generation tasks [9], [10], [11], [12], with the latest studies [13], [14], [15] confirming their impressive success in semantic segmentation tasks. Diffusion-based road extraction starts with random noise and aims to maximize the prior probability $p(x)$ that generates the mask $x$ under the implicit constraints of the input image. In this way, diffusion models can effectively adapt to the significantly complex backgrounds in remote sensing images

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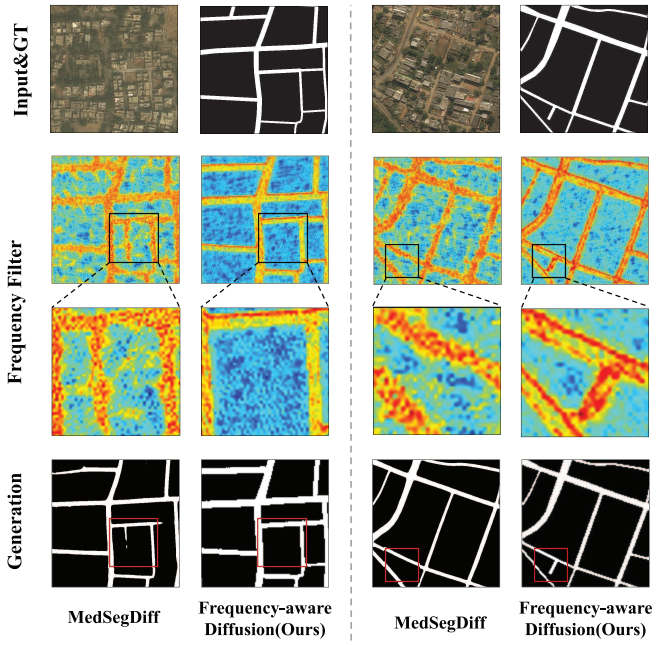


Fig. 1. Frequency filtering attention map and generated mask visualization. The brighter color indicates a higher attention score for roads. The Fourier transforms in MedSegDiff effectively remove high-frequency noise, but it still struggles to handle the complex frequency components in remote sensing images. In contrast, our proposed adaptive frequency-aware filtering shows much better performance.

and thoroughly capture the intricate details of road network bifurcations [16].

Although exploring diffusion-based generative paradigms for road extraction holds great potential, it still faces three major challenges. First, large-scale and high-resolution remote sensing images contain a significant amount of semantic features that are weakly correlated with road extraction. Diffusion models, as likelihood-based models, consume excessive computational resources to model these redundant details [17]. The additional gradient computations in the high-dimensional RGB image space substantially slow down the training and evaluation processes of the diffusion models [18]. Furthermore, the rich terrestrial features in remote sensing images are sensitive to noise within the diffusion models, especially for roads with large-scale differences. Poorly designed fusion modules can lead to a decline in segmentation performance. As illustrated in Fig. 1, although MedSegDiff [15] used Fourier transforms [19] to eliminate high-frequency noise, it is still insufficient for handling the complex frequency components in remote sensing images, leading to errors in the generated masks [20]. Finally, inadequate loss definitions are difficult to match with the high complexity of remote sensing image segmentation tasks. The existing studies [14], [21] typically consider only the variational boundary loss proposed in the original DDPM [22] paper. Although [23] introduced cross-entropy (CE) loss to enhance mask generation, it still fails to directly model the uncertainty introduced by random noise.

To address the aforementioned issues, in this article, we propose a novel adaptive frequency-aware latent diffusion framework (AFLDiff) for road extraction. AFLDiff starts with encoders for modeling latent variables to the latent space. Following common practices [12], [17], we pretrain a VQ-VAE [24] encoder and decoder tailored for road masks, which projects the masks from the original pixel space into a lower dimensional representation space and performs the reverse process. Unlike previous works that only reconstruct the mask [25], [26], AFLDiff introduces an additional latent perceptual compression module (LPCM) to suppress background-related low-frequency details in high-resolution remote sensing images. To facilitate the efficient fusion of noise and pixel features, we design a learnable adaptive frequency-aware filtering module named AFFM. AFFM relies on a frequency attention matrix [27] to enhance effective frequencies within the noise features, while the filter bank [28] integrates multiple sets of adaptive low-pass convolutional filters and high-pass pooling filters to filter out irrelevant frequency components during feature fusion. With a cross-attention mechanism, the conditioning latent variables are integrated into the noisy features, refining the generation of the denoised mask step by step. In addition, we introduce Gaussian encoders to model the probability distribution of the predicted mask and the noisy ground truth in the latent space, and Kullback–Leibler [29] divergence loss is applied to penalize the uncertainty introduced by Gaussian noise. By integrating these modules within the diffusion model framework, our approach can adaptively filter out frequency components irrelevant to road feature learning under acceptable computational cost, fully leveraging the powerful pixel-level probabilistic modeling capabilities in remote sensing images with complex land-cover features.

The main contributions are summarized as follows.

1) We propose an AFLDiff, which effectively alleviates the core challenges of insufficient frequency filtering and prohibitive computational demands when applying diffusion models to road extraction tasks in remote sensing images.
2) We introduce a learnable AFFM, which enhances relevant portions of noisy features using a frequency attention matrix and combines adaptive dynamic low-pass and high-pass filters to eliminate disruptive frequency components.
3) We present a latent space encoding module coupled with perceptual compression named LPCM. The module relies on a spatial attention mechanism to suppress irrelevant features while using the VQ-VAE encoder to model latent variables, further reducing the computational burden of diffusion models when applied to high-resolution remote sensing imagery.
4) AFLDiff achieves consistent state-of-the-art (SOTA) performance on three public datasets, and cross-dataset validation experiments further substantiate its exceptional robustness and generalization capabilities.

## II. Related Works

### A. Road Extraction

Traditional road extraction methods typically use manually designed texture features [30] or geometric features [31] and

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define specific matching criteria to detect roads. Conventional machine learning methods, such as support vector machines [32], random forests [33], and k-means clustering [34], have leveraged extensive datasets effectively. However, the acquired features often lacked the depth necessary to achieve the desired level of accuracy.

With the rapid development of artificial intelligence, road extraction from high-resolution remote sensing images using deep learning models mainly follows two approaches: segmentation-based and graph-based. Segmentation-based models treat road extraction as a pixel-level classification task, aiming to determine whether each pixel belongs to a road or not. Numerous CNN-based architectures such as SegNet [35], U-Net [36], DeepLab V3+ [37], and PSPNet [38] have been gradually used for learning road feature representations. Zhou et al. [39] proposed D-LinkNet based on the LinkNet [40] structure with added dilated convolution layers, which achieved the SOTA performance on the DeepGlobe dataset. CoANet [41] jointly learned segmentation and pairwise dependencies for road extraction, resulting in impressive extraction outcomes. While convolution-based models are adept at capturing local information, their global understanding is limited by the receptive field size, making it challenging to preserve the topological integrity of the extracted road network. Benefiting from their unique self-attention mechanism and global modeling capabilities, various transformer-based models have been progressively applied to road extraction tasks in remote sensing images. BDTNet [42] used a bidirectional transformer architecture to capture contextual road information at different scales. Liu et al. [43] combined the transformer with spatial and channel separable convolutions, further enhancing the model's feature representation capabilities and resulting in improved performance in capturing long-range information and global context. In addition, Ge et al. [44] introduced a U-shaped Swin Transformer architecture capable of handling both high-level and low-level feature information simultaneously. However, these methods primarily rely on the global perception capabilities of transformers for segmentation, which can be disrupted by the extensive occlusions present in remote sensing images. During the self-attention process, occluded road pixels may not receive attention equivalent to that of unobstructed road pixels, leading to fragmented road extraction results.

Graph-based methods have been proposed to preserve the topological integrity of the extracted road network, aiming to construct a road network graph that corresponds to the original image. SPIN Road Mapper [45] introduced a spatial and interaction space graph inference module, which combines graph convolution results in dual spaces to establish dependencies between road segments, enhancing the understanding of relationships between different areas. RoadTracer [46] proposed a method that iteratively constructs a graph by predicting vertices to detect road networks in remote sensing images. It uses a decision network to determine, based on a threshold, whether a candidate vertex should be added to the existing road network, resulting in a well-connected network structure. In [47], a heatmap representing the probability distribution of neighboring vertices for a given vertex was predicted, and local peaks were extracted as the predicted neighboring vertices, leading to improved directional prediction outcomes. Graph-based methods generally ensure the completeness of the global topological structure. However, when dealing with scenes that involve locally complex branching details, such as closed-loop overpasses, if an error occurs in vertex prediction or in the inference of dependencies for a particular area, the error may propagate during the iterative process, resulting in lower overall accuracy.

In addition, TPEGAN [7] proposed a framework for road extraction that enhances pixelwise segmentation and graph structure inference through pixel enhancement, effectively improving both accuracy and connectivity benchmarks. Nevertheless, it still fails to fully use the structural information within the mask, making it difficult to handle road branching details. Moreover, similar to other GAN-based methods, its adversarial training strategy is prone to mode collapse.

### B. Diffusion Models for Image Segmentation

Diffusion models [22] recover samples from noise using parameterized Markov chains and achieve excellent performance in various conditional and unconditional image generation tasks.

SegDiff [48] was the pioneering work that applied diffusion models to image segmentation problems. Since then, several studies [15], [49], [50] have worked to improve the performance of diffusion models for various downstream tasks in semantic segmentation. DFormer [14] leveraged a denoising perspective on universal image segmentation, introducing specific Gaussian noise to ground truth and training a model to predict denoising masks. Rahman et al. [50] used random sampling to generate the distribution of segmentation masks using multiple labels in combination with the randomness of the diffusion model implementing an advanced fuzzy segmentation network. Gu et al. [49] proposed DiffusionInst to represent instances as instance-aware filters and formulated instance segmentation as a noise-to-filter denoising process. The aforementioned methods apply image conditions at each timestep of the diffusion model to learn segmentation mappings and combine them with the segmentation mask of the current step. However, simply combining the feature-level information with the denoised mask of the current step may lead to incompatibility, making it crucial to filter out the frequency components that interfere with feature learning.

MedSegDiff [15] was the first to propose eliminating high-frequency noise in incomplete denoised masks in the Fourier space, effectively fusing the segmentation mask of the current step with the image prior, achieving remarkable success in medical image segmentation. However, the frequency components in remote sensing images are more complex, and Fourier transforms are insufficient to eliminate all the parts that interfere with feature learning. In this work, we bridge the gap between the noisy semantic features of the diffusion model and the encoded pixel features through adaptive feature filtering to enhance the performance for road extraction in remote sensing.

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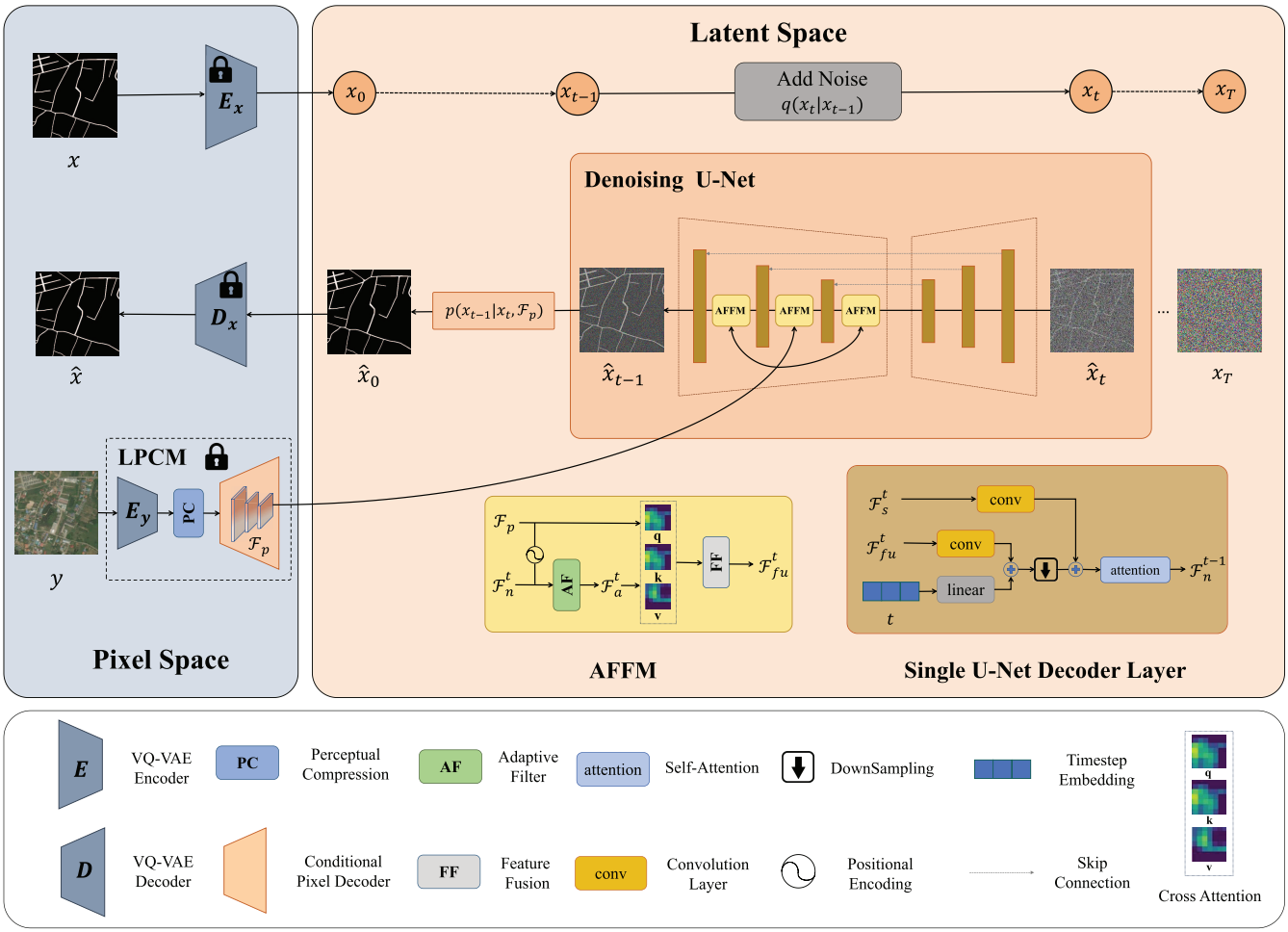


Fig. 2. Overall structure of AFLDiff. In the pixel space, the ground-truth mask $x$ is projected into the latent space as low-dimensional variables through the pretrained VQ-VAE encoder, and LPCM is designed to model multiscale latent pixel features $\mathcal{F}_p$ after perceptual compression for the original image $y$. In the latent space, the denoising U-Net relies on AFFM to adaptively filter the disruptive frequency components in the noisy features $\mathcal{F}_n$. The results are then fused with $\mathcal{F}_p$ using cross-attention mechanism to obtain $\mathcal{F}_{\text{fu}}$. After sufficient timesteps, a fully denoised mask is generated and inversely projected using the VQ-VAE decoder.

## III. Methods

### A. Overall Architecture

The overall architecture of AFLDiff is depicted in Fig. 2, comprising two main parts: the pixel space and the latent space. Within the pixel space, a pretrained VQ-VAE encoder $E_x$ and VQ-VAE decoder $D_x$ are integrated for latent space projection and inverse projection transformations. Their purpose is to model the road masks $x$ as latent variables $x_0$ in a low-dimensional representation space, which serves as the initial input for denoising U-Net. Once the denoising process is finished, the results $\hat{x}_0$ are projected back into the pixel space to produce the final prediction $\hat{x}$. Furthermore, a pretrained LPCM is designed for the original images $y$. LPCM models latent variables $z$ while coupling perceptual compression to suppress features in high-resolution remote sensing images that are weakly associated with roads. The reconstructed latent variables $y_R$ are then represented as multiscale pixel features $\mathcal{F}_p$ via the pixel decoder.

The core of the latent space is a diffusion model based on a denoising U-Net that frames road extraction as a mask-denoising task. In the forward process, random Gaussian noise is incrementally added to $x_0$ to obtain the noisy ground-truth mask set $x_t$. During the reverse process, the diffusion model predicts the added noise at each timestep, eventually generating the final denoised mask $\hat{x}_0$. At a fixed timestep $t$, the AFFM filters out disruptive components from the noisy features $\mathcal{F}_n^t$, while enhancing relevant high-frequency and low-frequency information to obtain the aggregated features $\mathcal{F}_a^t$. The frequency-rich $\mathcal{F}_{\text{agg}}^t$ is then fused with the multiscale pixel features $\mathcal{F}_p$ via a cross-attention mechanism, facilitating effective guided feature learning for segmentation. The fused result is represented as $\mathcal{F}_{\text{fu}}^t$.

In the denoising U-Net decoder, each layer receives the features $\mathcal{F}_s^t$ from the skip connections. After embedding the timestep $t$, the fused features $\mathcal{F}_{\text{fu}}^t$ are processed through convolutional transformations and self-attention mechanism and are then regarded as noisy features for the next timestep $\mathcal{F}_n^{t-1}$, progressively refining the generated road masks.

### B. Latent Perceptual Compression Module

Within high-resolution remote sensing images, the background is complex and varied, containing numerous land features. The pattern coverage behavior of diffusion models

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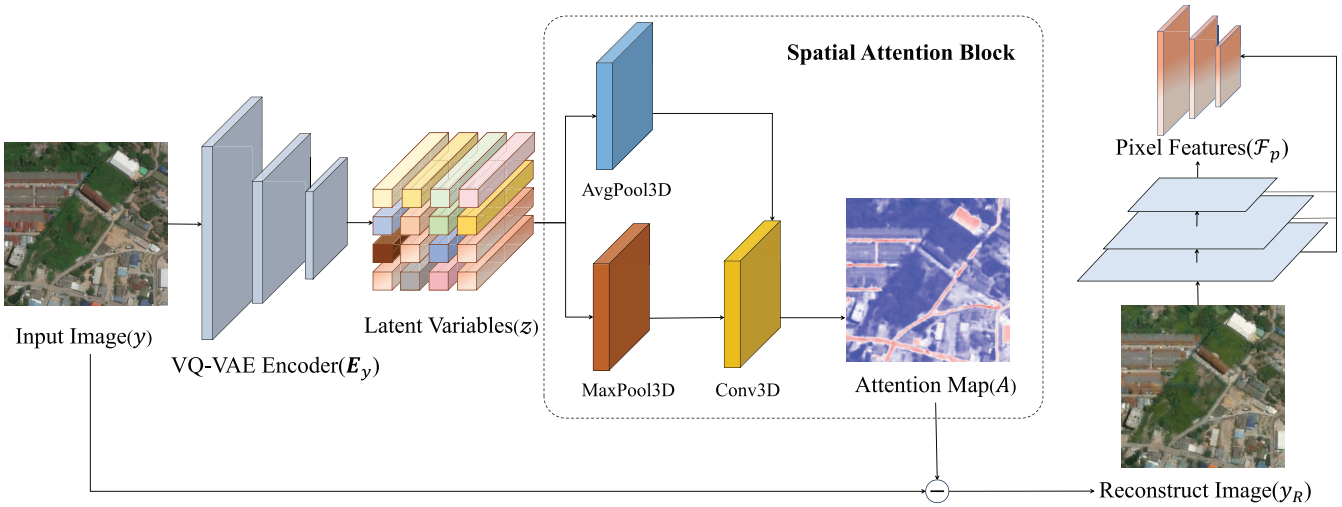


Fig. 3. Structure of LPCM. Latent variables $z$ projected by the VQ-VAE encoder are enhanced with spatial attention to generate a reconstructed image $y_R$ focusing on roads, which is then modeled as multiscale pixel features $\mathcal{F}_p$ using a feature pyramid decoder.

may lead to substantial computational resource consumption on features that are weakly correlated with the road extraction task [17]. Therefore, it is crucial to implement perceptual compression on the input images. Our designed LPCM, as shown in Fig. 3, consists of a VQ-VAE encoder [24], a spatial attention block [51], and a multiscale feature pyramid decoder. LPCM projects the input images to low-dimensional latent variables and uses a cross-attention mechanism to compress regions of the images with weak associations to road features, ultimately producing multiscale pixel features after decoding.

Specifically, for an input image $y \in \mathcal{R}^{H*W*3}$, we use a pretrained VQ-VAE encoder to perform dimensionality reduction into the space defined by vector discretization. We apply a soft assignment of instances $\{i = 1, \ldots, K\}$ for each pixel using a codebook, resulting in latent variables $z \in \mathcal{R}^{h*w*C}$ (where $h = H/m$ and $w = W/m$, with $m$ being the downsampling factor and $C$ being the hidden dimensions). This process can be viewed as modeling a posterior probability distribution $q(z|y)$, which is used for mapping the input image from the high-dimensional pixel space to the low-dimensional latent space.

For the latent variable $z$, a spatial attention mechanism is used to obtain a road feature attention map. Average pooling and max pooling calculations aggregate the spatial information of the feature maps, where average pooling maintains environmental context through global context awareness, and max pooling computes local feature saliency to highlight critical edge features. After concatenating along the channel dimension, a convolution calculation is performed to generate the spatial attention weight map $A$. The calculation process is defined as follows:

$$A = \sigma_{3\times3}\left(\left[\text{AvgPool}\,(z) \oplus \text{MaxPool}\,(z)\right]\right). \tag{1}$$

In the equation, $\sigma_{3\times3}$ represents a convolution layer with a kernel size of $3 \times 3$; AvgPool and MaxPool denote the average pooling and max pooling calculations, respectively; and $\oplus$ means concatenation operations along the channel dimension.

The feature reconstructed after perceptual compression is obtained by multiplying $y$ and $A$ followed by a subtraction operation. Following [52], to further control the computational burden caused by multiple computations of high-resolution feature maps, we use a pixel decoder with a multiscale feature pyramid decoder. Specifically, the feature pyramid consists of three layers, with the output feature map resolutions being 1/16, 1/4, and 1 of the original image, respectively. The feature maps, arranged from coarse to fine scales, are designed to progressively preserve global road network information, road orientation information, and road boundary details, respectively. Through multilevel sampling, the decoded pixel features fuse high-level semantics with low-level semantics, effectively addressing the large-scale variations in roads in high-resolution remote sensing images. The pixel features extracted by the pixel decoder are represented as $\mathcal{F}_p^i \in R^{(h/s_i)*(w/s_i)*C}$, where $s_i = 2^i (i = 0, 1, 2)$ represents the stride mapped to the feature map.

### C. Adaptive Frequency-Aware Filter Module

Semantic segmentation is a challenging pixel-level classification task, especially in remote sensing images. Zhong et al. [53] demonstrated that frequency representation can serve as a paradigm for capturing interclass differences, offering valuable guidance for semantic segmentation. The noisy features obtained during the diffusion process contain noise at various frequencies, and directly combining these features with semantic features from the pixel encoder can lead to confusion during feature fusion due to domain gaps [54]. To mitigate the negative impact of redundant frequency components on feature learning, we design an AFFM. As shown in Fig. 4, AFFM consists of two branches, one for adaptive frequency-aware filtering and another for feature fusion. The adaptive frequency-aware filtering branch is responsible for filtering out effective high-frequency and low-frequency components from the noisy features, while the feature fusion branch uses

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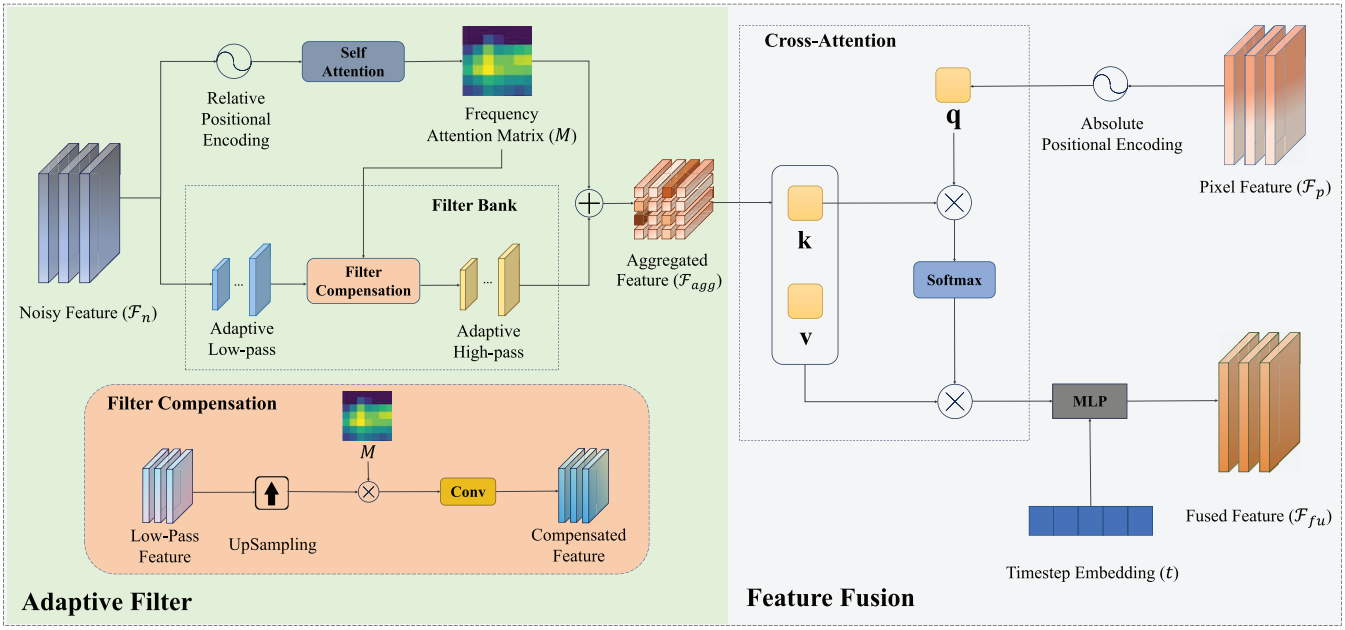


Fig. 4. Structure of AFFM. Adaptive filter branch relies on a frequency attention matrix and a filter bank integrated with adaptive high-pass and low-pass filters to filter out components weakly associated with roads in $\mathcal{F}_n$. Feature fusion branch uses the cross-attention mechanism to obtain $\mathcal{F}_{\text{fu}}$.

a cross-attention mechanism to fuse the noisy features with pixel features at multiple scales.

*1) Adaptive Filter:* At each denoising timestep, the noisy features $\mathcal{F}_n$ contain pixel-level information from the images and road features from the masks integrated from the previous step, encompassing complex frequency components. Therefore, it is crucial to extract elements that contribute meaningfully to semantic understanding. The adaptive frequency-aware filtering begins with an attention-based approach aimed at extracting a frequency attention matrix $M_{i,j}$, which represents the weights of different frequency components. This process leverages self-attention applied to the noisy features. For $\mathcal{F}_n$, we introduce relative positional encoding [55] to assign a 2-D relative positional bias to each spatial position, and the encoding process is defined as follows:

$$\mathcal{F}_{\text{enc}}(x,y) = \mathcal{F}_n(x,y) + f\left(\sqrt{(x-x_0)^2+(y-y_0)^2}\right). \quad (2)$$

In the equation, $\mathcal{F}_{\text{enc}} \in \mathcal{R}^{h*w*C}$ represents positional encoded features; $f$ denotes the mapping function; and $(x_0, y_0)$ is the selected reference point.

$\mathcal{F}_{\text{enc}}$ passes through a 2-D convolutional layer with a kernel size of $(h, w)$ and is reshaped to $\mathcal{F}_{\text{enc}} \in \mathcal{R}^{C*C}$. Through learnable transformations, $\mathcal{F}_{\text{enc}}$ is mapped into key $\boldsymbol{K} = \mathcal{F}_{\text{enc}} W^k$ and query $\boldsymbol{Q} = \mathcal{F}_{\text{enc}} W^q$. The product of the keys and queries is computed via a linear layer, and a softmax operation normalizes the keys across the frequency components, producing a matrix that assigns weights to each component

$$M_{i,j} = \text{Softmax}\left(\frac{e^{\boldsymbol{Q}\boldsymbol{K}^{\text{T}}}}{\sqrt{d_{\boldsymbol{K}}}}\right) = \text{Softmax}\left(\frac{e^{\boldsymbol{q}_{i,:}\boldsymbol{k}_{:,j}^{\text{T}}}}{\sqrt{C}}\right). \quad (3)$$

In the equation, $\boldsymbol{q}_{i,:}$ and $\boldsymbol{k}_{:,j}$ represent the $i$th row of the key matrix and the $j$th column of the query matrix, respectively.

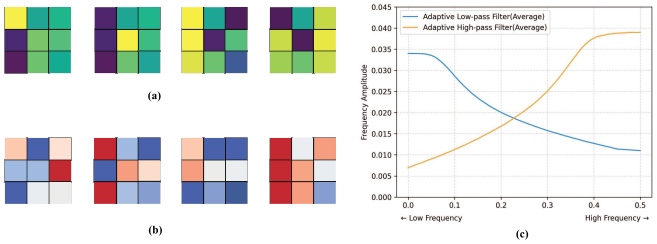


Fig. 5. Frequency analysis of filter bank. (a) Adaptive depthwise convolutional kernels visualization. (b) Adaptive average pooling kernels' visualization. (c) Filtering effect on different frequency components of the filter bank trained on the DeepGlobe dataset.

On this basis, another learnable transformation is applied to convert the encoded features into value $\boldsymbol{V} = \mathcal{F}_{\text{enc}} W^v$. The product of values and the frequency attention matrix represents the frequency components after enhancement, forming the frequency-aware feature expressed as

$$\mathcal{F}_{\text{fa}} = \boldsymbol{M}\boldsymbol{V}. \quad (4)$$

In the equation, $M$ represents the frequency attention matrix, and $\boldsymbol{V}$ denotes the query vector transformed from the encoded features.

For high-resolution remote sensing images, the low-frequency components represent most of the flat background information, while the high-frequency components retain important terrain feature information. Traditional single-pass filtering methods may lose some critical semantic features from the original image [56]. Inspired by [27], we use a filter bank [28] that combines dynamic adaptive low-pass and adaptive high-pass filters for frequency filtering. The adaptive low-pass filter consists of a set of $3 \times 3$ average pooling kernels, while the adaptive high-pass filters include a set of $3 \times 3$ depthwise convolution kernels. Unlike conven-

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tional fixed-parameter filtering algorithms, the adaptive low-pass/high-pass filters dynamically optimize their frequency responses via task-driven parameter adaptation, achieving context-aware noise suppression in complex remote sensing environments where heterogeneous frequency components coexist. The analysis of the filter bank's frequency filtering effect is shown in Fig. 5. The adaptive high-pass and low-pass filters are sensitive to the high-frequency and low-frequency components of the noise features, respectively. Due to the inherent patterns of the convolution and pooling layers, they have a significant ability to capture the adapted frequency components.

A filter compensation module is incorporated between the adaptive low-pass and high-pass filters. The module first performs upsampling to expand the dimensionality of the features after low-pass filtering, and then computes the compensated features by applying a dot product with the frequency attention matrix. Finally, the dimensionality of the compensated features is adjusted through a convolutional layer before being output. The purpose of the filter compensation design is to mitigate the significant loss of frequency information caused by insufficiently trained low-pass filters in the early stages of training. As training progresses, the filters and the frequency-aware attention matrix become aligned, ensuring the effectiveness of the filtering operation. The filtered features are represented as

$$\mathcal{F}_{\text{fi}} = \tau_{s_1} \cdots \tau_{s_n} \left( \text{Conv} \left( M \otimes \text{up} \left( \sigma_{k_1} \cdots \sigma_{k_n} \left( \mathcal{F}_n \right) \right) \right) \right). \tag{5}$$

In the equation, $\sigma_{k_n}$ represents an average pooling kernel $k_n$; $\tau_{s_n}$ stands for a depthwise convolution kernel $s_n$; Conv denotes convolutional computation; up is the upsampling operation based on bilinear interpolation; and $\otimes$ represents the matrix dot product operation.

The final filtered frequency-aware feature representation is obtained by aggregating the results from the two branches, expressed as

$$\mathcal{F}_{\text{agg}} = \alpha F_{\text{fa}} + \boldsymbol{\beta} \mathcal{F}_{\text{fi}}. \tag{6}$$

In the equation, $\alpha$ and $\beta$ represent weights used to limit the frequency contribution of the branch features during aggregation.

At different denoising timesteps, the comparison of feature heatmaps before and after adaptive filtering is shown in Fig. 6. The filtered features include channels derived from both low-pass and high-pass filtering and evidently suppress the noisy frequency components that are widely present in the noisy features. In the heatmaps across all the channels, the feature intensities of roads and other semantics show clear distinctions, highlighting the effectiveness of adaptive filtering.

*2) Feature Fusion:* For the aggregated features from the adaptive frequency-aware filtering branch $\mathcal{F}_{\text{agg}}$ and the pixel features from the $i$th layer of the pyramid $\mathcal{F}_p^i$, it is essential to integrate the preserved effective low-frequency semantic information and high-frequency detail information with the pixel-level features. Based on this, the denoising U-Net can learn the interaction between noise and semantic features, thereby achieving a stronger representational capability.

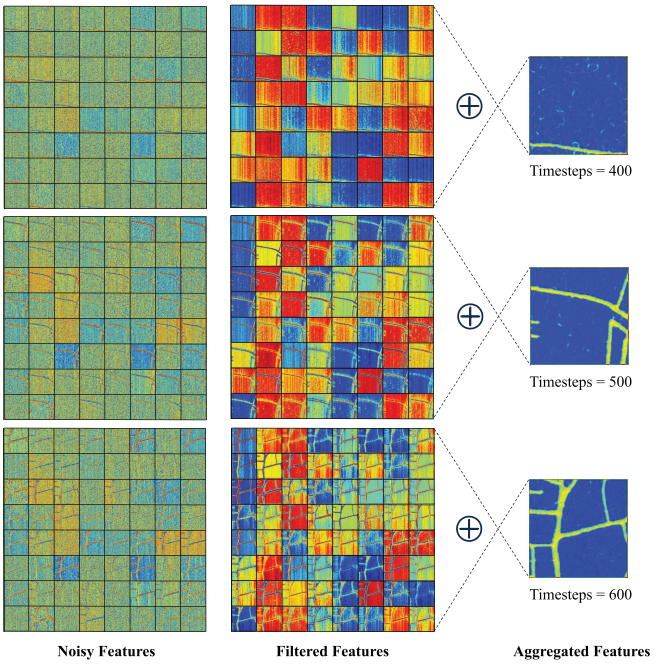


Fig. 6. Feature heatmap comparison before and after adaptive filter. The brighter colors indicate higher feature intensity.

The feature fusion process relies on the cross-attention computation between $\mathcal{F}_{\text{agg}}$ and $\mathcal{F}_p^i$. Specifically, $\mathcal{F}_{\text{agg}}$ is transformed into key vector $\boldsymbol{K}$ and value vector $\boldsymbol{V}$ using learnable transformation matrices $W^k$ and $W^v$, while $\mathcal{F}_p^i$ is first embedded with absolute positional encoding and then converted into query vectors $\boldsymbol{Q}$ using the learnable matrix $W^q$. The similarity between $\boldsymbol{Q}$ and $\boldsymbol{K}$ is computed to generate an attention distribution, which is used to perform a weighted sum of $\boldsymbol{V}$, resulting in the fused features. The computation involved in this process is represented by the following equation:

$$\boldsymbol{K} = \left( \mathcal{F}_{\text{agg}} \right) W^k, \quad \boldsymbol{Q} = \left( \mathcal{F}_{\text{agg}} \right) W^q, \quad \boldsymbol{V} = \text{PE} \left( \mathcal{F}_p^i \right) W^v \tag{7}$$

$$\mathcal{F}_{\text{att}} = \text{Softmax} \left( \frac{\boldsymbol{Q}\boldsymbol{K}^{\text{T}}}{\sqrt{C}} \right) \boldsymbol{V}. \tag{8}$$

In the equation, $W^k$, $W^q$, and $W^v$ are all learnable parameter matrices, and $C$ represents the embedding dimension.

We follow the classic diffusion implementation [22] to apply AdaIN normalization to the attention features and embed the time step information, allowing the fusion module to adapt to the step information. Finally, a multilayer perceptron is used for further refinement. The resulting final fused feature representation is

$$\mathcal{F}_{\text{fu}} = \text{MLP} \left( f_{\text{norm}} \left( \mathcal{F}_{\text{att}} \right) + \sin \left( \frac{2\pi i}{T} \cdot t \right) \right). \tag{9}$$

In the equation, $f_{\text{norm}}$ represents the AdaIN normalization; $i$ denotes the pyramid feature layer number; $T$ indicates the total number of sampling time steps; and $t$ represents the current time step.

The fused features obtained from the current AFFM will be treated as the recovered noisy features and will be passed for fusion with deeper pyramid features. This ensures that at the same time step, different scales of pixel features progressively

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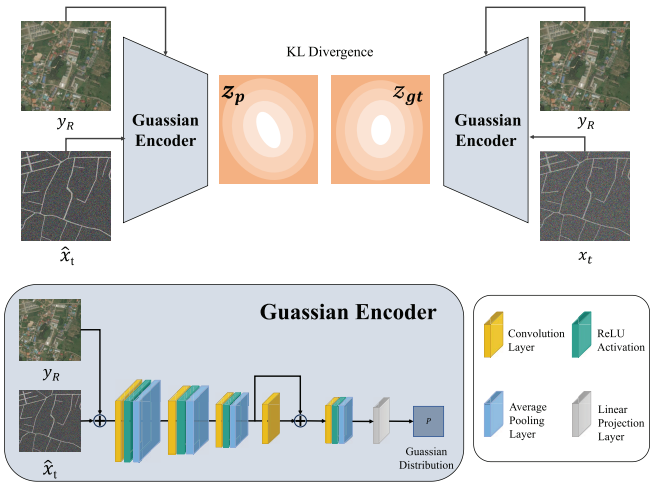


Fig. 7. Modeling of Gaussian probability distributions and discrepancy quantification. The Gaussian encoder models the Gaussian probability distributions corresponding to the latent variables $z_p$ jointly formed by the predicted denoised mask $\hat{x}_t$ and the reconstructed image $y_R$, with KL divergence being the discrepancy loss.

refine the representation of the noisy features, guiding the diffusion model to stably generate accurate road masks.

### D. Gaussian Encoder and Loss Definitions

Denoising U-Net, the core part of AFLDiff, is trained in a latent space. At time step $t$, the predicted denoising mask is sampled from a Gaussian probability distribution, which fundamentally determines the accuracy of the prediction results. Therefore, quantifying the discrepancy between this distribution and the true distribution is of great importance.

The latent variables of road masks modeled by the pretrained VQ-VAE inherently align with the discrete sampling of a Gaussian mixture distribution. By introducing controlled Gaussian noise, the resulting distribution still remains analytically tractable under Gaussian modeling according to the Gaussian superposition theorem. To this end, we design two Gaussian encoders, as shown in Fig. 7. With the reconstructed image used as a condition, the encoders model the predicted mask at the current time step and the true mask with added noise as Gaussian probability distributions, incorporating the Kullback–Leibler divergence [29] between the two distributions as a penalty term.

Both the Gaussian encoders have identical diagonal covariance matrices, each consisting of a stack of four convolutional layers of $3 \times 3$, with each layer containing a ReLU activation and an average pooling layer of $2 \times 2$ for downsampling. A residual connection is incorporated in the middle of the encoder to ensure effective feature extraction. The Gaussian probability distribution, formed by the parameters predicted through the linear layer, is defined as

$$\begin{aligned} P\left(z_p|y_R, \hat{x}_t\right) &\sim N\left(\mu\left(y_R, \hat{x}_t\right), \sigma\left(y_R, \hat{x}_t\right)\right) \\ P\left(z_{\text{gt}}|y_R, x_t\right) &\sim N\left(\mu\left(y_R, x_t\right), \sigma\left(y_R, x_t\right)\right). \end{aligned} \tag{10}$$

In the equation, $z_p$ and $z_{\text{gt}}$ represent the latent variables modeled by Gaussian encoder, and $P$ denotes the Gaussian probability distribution formed by the predicted mean $\mu(y_R, \widehat{x}_t) \in R^C$ and variance $\sigma(y_R, \widehat{x}_t) \in R^{C\times C}$.

We use the KL divergence loss to measure the discrepancy loss between these two probability distributions, which is defined as

$$\mathcal{L}_{\text{disk}} = \mathbb{D}_{\text{KL}}\left[P\left(z_p\right)\left|\left|P\left(z_{\text{gt}}\right)\right.\right]\right. . \tag{11}$$

In addition, we also incorporate the CE loss between the predicted mask and the ground truth, as well as the mean square error loss between the predicted noise and the true noise. These two losses are defined as follows:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[x_t \cdot \log\left(\hat{x}_t\right) + \left(1 - x_t\right) \cdot \log\left(1 - \hat{x}_t\right)\right] \tag{12}$$

$$\mathcal{L}_{\text{mse}} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\epsilon}_t - \epsilon_t\right)^2 . \tag{13}$$

In the equations, $N$ represents the total number of pixels; $t$ is the selected timestep; $\hat{x}_t$ and $x_t$ denote the predicted mask and the ground-truth mask at current timestep, respectively; and $\hat{\epsilon}_t$ and $\epsilon_t$ are the predicted noise and the added Gaussian noise at current timestep, respectively.

The total loss during training is defined as the sum of these three loss functions, represented as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{disk}} + \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{mse}}. \tag{14}$$

### E. Training and Inference

During the training process, at a randomly selected timestep, the road mask is projected into the latent space using a pretrained VQ-VAE encoder and augmented with a fixed amount of Gaussian noise learned during the forward process, serving as the noisy mask ground truth for the current timestep. At the same time, the original image is perceptually compressed using LPCM, modeled into multiscale pixel features, and projected into the latent space as an implicit conditional constraint. The denoising U-Net iteratively learns the prior probability distribution of road features within the mask over timesteps. It uses the AFFM to filter out interference components from the noisy mask and enhance effective features, which are then fused with pixel features through a cross-attention mechanism. The fused features serve as input for the next timestep, progressively refining the generation of a fully denoised road mask. At the end of the iterations, the generated mask is inversely projected back to pixel space using the pretrained VQ-VAE decoder. The training loss consists of three parts, which, respectively, regularize the predicted mask, the predicted noise, and the probabilistic distribution of the predicted latent variables modeled by the Gaussian encoder.

During the inference process, the denoising U-Net starts with randomly sampled noise and iteratively performs denoising under the guidance of the reconstructed latent pixel features from the original image. This process ultimately generates the predicted denoised latent variables, which are then inversely projected to the pixel space using the pretrained VQ-VAE decoder to obtain the predicted road mask.

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## IV. Datasets and Performance Metrics

### A. Datasets

*DeepGlobe [57]* consists of images captured over Thailand, Indonesia, and India, with a resolution of 50 cm/pixel, and the size of each image is 1024 × 1024. In the experiments, we randomly divide the labeled images into 4980 for training, 623 for validation, and 623 for testing. For the training set and validation set, the images of size 1024 × 1024 are cropped to 512 × 512 image patches with overlapping windows of 256. For the test set, the images are cropped without overlapping pixels. After cropping, the three sets are expanded to 42 030, 5607, and 2492 images, respectively.

*Massachusetts [58]* comprises 1171 labeled satellite images over a diverse range of urban, suburban, and rural regions. It spans an area of approximately 2600 square kilometers with a resolution of 120 cm/pixel, and each image has a size of 1500 × 1500 pixels. The dataset includes 1108 images for training, 14 for validation, and 49 for testing. Following similar operations, the processed Massachusetts dataset contains 27 700 patches for training, 350 for validation, and 441 for testing, all at a size of 512 × 512.

*CHN6-CUG [59]* was obtained from Google Earth, providing 4511 annotated images from six different cities in China with a size of 512 × 512. The dataset includes multiple types of roads such as railways, highways, urban streets, and rural pathways. Within the dataset, the images are divided into 3608 for training, 602 for validation, and 301 for testing.

### B. Performance Metrics

To quantitatively evaluate the accuracy performance of the proposed AFLDiff on road extraction tasks, we include widely used evaluation metrics: precision, recall, dice score, mean intersection over union (mIoU), and road intersection over union (rIoU). Road extraction is a binary segmentation problem, where the classification of pixels in the prediction results includes the following four categories: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). Based on this, the formulas for calculating these evaluation metrics are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{15}$$

$$\text{dice} = F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{16}$$

$$\text{mIoU} = \frac{1}{2} \cdot \left( \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} + \frac{\text{TN}}{\text{TN}+\text{FP}+\text{FN}} \right) \tag{17}$$

$$\text{rIoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}}. \tag{18}$$

In addition, to assess computational efficiency, we incorporate the model parameters (Params) and giga floating-point operations per second (GFLOPS). GFLOPS metric is measured by inferring 256 × 256 images on an Nvidia A100 graphics processing unit.

## V. Experimental Analysis

### A. Implementation Details

Our proposed AFLDiff is implemented by PyTorch 1.10 on a single NVIDIA A100 GPU. The VQ-VAE encoder and decoder are trained for 250 epochs with a batch size of 16, optimized using the Adam optimizer with an initial learning rate of 0.001. The learning rate is adjusted using the StepLR method, with a step size of 30 and a decay factor of 0.5. The denoising U-Net is trained for 1000 epochs with a batch size of 32 and the total timesteps set to 1000. It is optimized using the AdamW optimizer with an initial learning rate of 0.0004 and a weight decay coefficient of 1e-6. Similarly, the learning rate is adjusted using the StepLR method, with a step size of 50 and a decay factor of 0.9. In addition, the training process is accelerated using the accelerator library.

During training, a data augmentation strategy has been used to expand the training set. Image transformations, including random rotations of 90°, 180°, and 270°, as well as random flips along the vertical and horizontal axes, are applied to all the datasets. Notably, due to the substantial memory demands caused by timestep accumulation in the diffusion model [67], the downsampling factor in the reported AFLDiff is set to $m = 2$, resulting in a latent variable size of 256 × 256.

### B. Comparison to SOTA Methods

To the best of our knowledge, AFLDiff stands out as the pioneering unified framework that incorporates the diffusion model into road extraction tasks in high-resolution remote sensing images. To validate the effectiveness of AFLDiff in road extraction tasks, we select several representative methods as benchmarks, including CNN-based methods U-Net [36] and D-LinkNet [39], transformer-based method TransUNet [60], graph-based method SPIN Road Mapper [45], and diffusion-based general segmentation method MedSegDiff [15]. In addition, we include recent studies for comparative experiments, such as Attention-UNet [61], SegFormer [62], SegDiff [48], RADANet [6], AFU-Net [63], TPEGAN [7], AFCNet [64], FERDNet [65], RoadMamba [66], and DENet [68].

*1) Qualitative Results:* The predicted road masks of AFLDiff and the comparative models on the three public datasets are illustrated in Fig. 8. Our proposed AFLDiff excels in capturing fine branching details and maintaining connectivity with the main roads compared with the existing methods. For instance, in the segmentation results shown in row (c), comparative methods exhibit varying degrees of road loss, while AFLDiff produces satisfying results. In addition, the robust representation capabilities learned from fusing noise and multiscale semantic features in AFLDiff allow it to adapt well to dense environmental backgrounds. Notably, in rows (a) and (e), where the roads are partially obscured by buildings or visually similar to surrounding structures, AFLDiff still manages to effectively distinguish roads from other elements in the scene, using its powerful probabilistic sampling capability. For the more extensive urban highways depicted in rows (g) and (h), AFLDiff demonstrates superior capability in accurately delineating intricate ring-shaped structures. In the visualization results, diffusion-based methods such as MedSegDiff and AFLDiff exhibit low-resolution segmentation results. This is due to the downsampling applied to road masks during the VQ-VAE encoding process, as mentioned earlier, to accommodate computational resource limitations.

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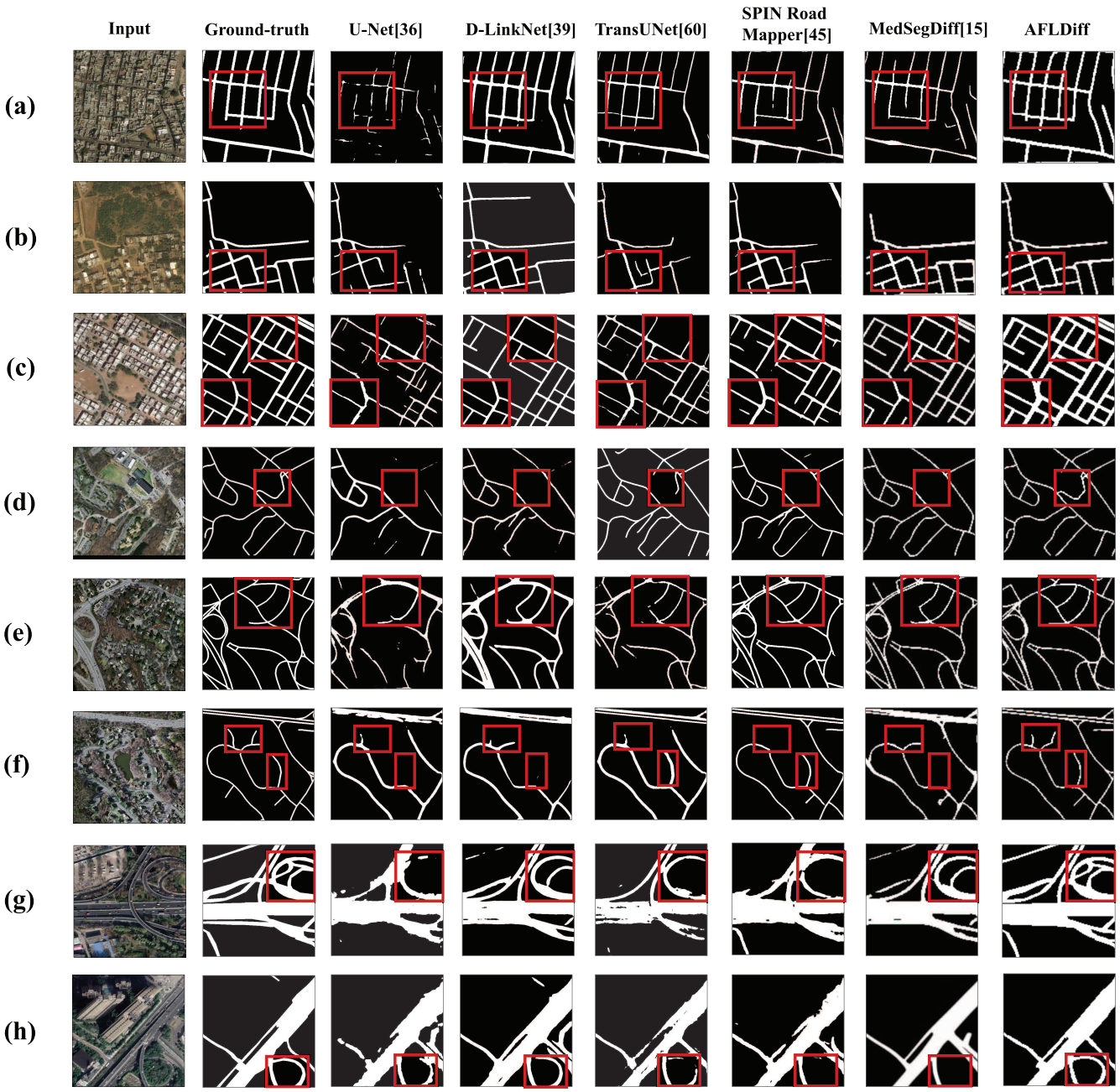


Fig. 8. Visualization results of AFLDiff and comparison benchmarks. (a)–(c) Results from the DeepGlobe dataset. (d)–(f) Results from the Massachusetts dataset. (g) and (h) Results from the CHN6-CUG dataset. Red boxes mark the areas where AFLDiff outperforms other methods.

*2) Quantitative Results:* The quantitative results of AFLDiff and the comparative methods on two datasets are presented in Tables I and II. AFLDiff basically consistently achieves SOTA performance across all three datasets.

On the DeepGlobe dataset, AFLDiff achieves a maximum mIoU of 82.01%, rIoU of 76.25%, and dice score of 85.96%. It surpasses the recent existing methods by 0.20% in mIoU, 2.57% in rIoU, and 1.12% in dice score. These results indicate that AFLDiff can accurately extract roads from high-resolution remote sensing images with fewer misclassifications and omissions. Compared with MedSegDiff, which uses diffusion models for medical image segmentation, our improvements enhance mIoU by 4.63%, rIoU by 4.97%, and dice score by 2.73%. This indicates that the improvement from Fourier spectral transformation to adaptive frequency-aware filtering, along with the introduction of KL loss, significantly enhances the fine-grained generation of road masks.

On the Massachusetts and CHN6-CUG datasets, AFLDiff also achieves significant performance improvements. Compared with graph-based methods such as SPIN Road Mapper, AFLDiff still exhibits improvements of 2.68% and 2.99% in rIoU on the two datasets, respectively, indicating high connectivity and accuracy of the predicted road masks. In contrast to transformer-based methods such as TransUNet and SegFormer,

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TABLE I

QUANTITATIVE EVALUATIONS OF AFLDIFF WITH SOTA METHODS ON THE DEEPGLOBE DATASET. THE BEST RESULTS ARE HIGHLIGHTED, THE SECOND-BEST RESULTS ARE UNDERLINED

| Category | Method | Precision(%)↑ | Recall(%)↑ | mIoU(%)↑ | rIoU(%)↑ | dice(%)↑ | Params(M) ↓ | GFLOPS ↓ (256*256) |
|---|---|---|---|---|---|---|---|---|
| General | U-Net$_{2015'}$ [36] | 74.26 | 68.17 | 59.36 | 55.52 | 71.07 | <u>21.74</u> | <u>19.61</u> |
| | TransUNet$_{2021'}$ [60] | 78.61 | 68.22 | 66.43 | 57.54 | 73.05 | 82.87 | 24.88 |
| | Attention-UNet$_{2022'}$ [61] | 79.89 | 74.57 | 70.31 | 62.79 | 77.14 | 34.88 | 41.51 |
| | SegFormer$_{2021'}$ [62] | 81.86 | 79.58 | 73.42 | 68.65 | 80.70 | 67.18 | 44.57 |
| | SegDiff$_{2021'}$ [48] | 83.72 | 81.22 | 73.87 | 70.41 | 82.45 | 39.37 | 78.84 |
| | MedSegDiff$_{2024'}$ [15] | 86.37 | 82.15 | 77.38 | 71.28 | 83.23 | 48.41 | 61.44 |
| Specialized | D-LinkNet$_{2018'}$ [39] | 81.33 | 74.29 | 68.52 | 63.47 | 77.65 | 31.10 | **16.40** |
| | SPIN Road Mapper$_{2022'}$ [45] | 84.14 | 82.50 | 72.89 | 69.02 | 83.31 | 58.04 | 32.59 |
| | AFU-Net$_{2023'}$ [63] | 87.86 | 81.70 | 74.14 | 73.42 | 84.67 | 110.04 | 23.92 |
| | TPEGAN$_{2024'}$ [7] | 86.17 | 82.49 | 77.19 | 71.84 | 84.29 | 102.71 | 45.91 |
| | AFCNet$_{2024'}$ [64] | <u>88.10</u> | 81.81 | 78.26 | <u>73.68</u> | <u>84.84</u> | 34.09 | 28.10 |
| | FERDNet$_{2025'}$ [65] | 83.40 | <u>83.10</u> | / | 72.70 | 84.20 | **4.10** | 27.33 |
| | RoadMamba$_{2025'}$ [66] | / | / | <u>81.81</u> | 67.33 | 80.48 | / | / |
| Proposed | AFLDiff($m$=2) | **89.43** | **83.62** | **82.01** | **76.25** | **85.96** | 42.21 | 41.35 |

TABLE II

QUANTITATIVE EVALUATIONS OF AFLDIFF WITH SOTA METHODS ON THE MASSACHUSETTS AND CHN6-CUG DATASETS. THE BEST RESULTS ARE HIGHLIGHTED, THE SECOND-BEST RESULTS ARE UNDERLINED

| Category | Method | Massachusetts | | | CHN6-CUG | | |
|---|---|---|---|---|---|---|---|
| | | mIoU(%)↑ | rIoU(%)↑ | dice(%)↑ | mIoU(%)↑ | rIoU(%)↑ | dice(%)↑ |
| General | U-Net$_{2015'}$ [36] | 76.39 | 49.91 | 66.59 | 74.50 | 56.71 | 72.28 |
| | TransUNet$_{2021'}$ [60] | 77.89 | 54.98 | 70.95 | 75.06 | 58.52 | 73.80 |
| | Attention-UNet$_{2022'}$ [61] | 78.13 | 58.79 | 72.35 | 74.64 | 57.88 | 73.49 |
| | SegFormer$_{2021'}$ [62] | 78.33 | 59.48 | 72.92 | 76.13 | 58.84 | 74.44 |
| | SegDiff$_{2021'}$ [48] | 79.02 | 58.25 | 72.81 | 77.20 | 59.05 | 75.36 |
| | MedSegDiff$_{2024'}$ [15] | 81.67 | 61.31 | 76.41 | 79.58 | 62.18 | 77.29 |
| Specialized | D-LinkNet$_{2018'}$ [39] | 77.23 | 53.10 | 69.37 | 79.61 | 61.79 | 76.21 |
| | SPIN Road Mapper$_{2022'}$ [45] | 80.74 | 63.16 | 76.41 | 79.53 | 62.81 | 76.35 |
| | RADANet$_{2023'}$ [6] | 81.65 | 61.11 | 75.86 | / | 60.43 | 75.34 |
| | AFU-Net$_{2023'}$ [63] | 80.32 | 61.28 | 75.99 | / | / | / |
| | TPEGAN$_{2024'}$ [7] | 81.72 | 60.02 | 73.43 | 79.60 | 62.56 | 76.72 |
| | AFCNet$_{2024'}$ [64] | <u>83.15</u> | 61.77 | 76.37 | / | / | / |
| | RoadMamba$_{2025'}$ [66] | 81.42 | 64.92 | 78.73 | 80.68 | 63.98 | 78.03 |
| | DENet$_{2025'}$ [68] | 81.74 | <u>65.76</u> | **79.35** | <u>80.94</u> | <u>64.63</u> | <u>78.51</u> |
| Proposed | AFLDiff($m$=2) | **85.28** | **65.84** | <u>78.90</u> | **82.45** | **65.80** | **78.79** |

AFLDiff demonstrates a considerable performance gap, further proving its competitive performance in road extraction tasks.

The efficiency comparison results of AFLDiff with other models are illustrated in Fig. 9. It can be seen that our proposed AFLDiff is generally positioned in the upper-middle range of the performance comparison chart, with model parameters roughly equal to the average of the comparison methods. This indicates that AFLDiff achieves a good balance between accuracy and computational efficiency. In addition, compared with other diffusion-based methods such as MedSegDiff, AFLDiff achieves a 12.81% reduction in parameter count and a 32.70% improvement in computational efficiency, demonstrating that LPCM's latent variables modeling ability contributes to the efficiency enhancement of AFLDiff.

### C. Ablation Study

In this section, we conduct ablation studies on modules, the diffusion sampling time steps, and loss items to validate the effectiveness of our design, and to explore the tradeoff between model accuracy and sampling efficiency.

*1) Ablation Studies on Modules:* AFLDiff consists of three essential module designs: LPCM, AFFM, and the Gaussian encoders incorporating KL loss. We start with a baseline model and progressively integrate these three modules to

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TABLE III
ABLATION STUDY RESULTS ON THE DEEPGLOBE AND MASSACHUSETTS DATASETS. THE BEST RESULTS ARE HIGHLIGHTED

| LPCM | AFFM | GE+KL Loss | DeepGlobe | | | Massachusetts | | | Params (M) | GFLOPS |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | mIoU(%) | rIoU(%) | dice(%) | mIoU(%) | rIoU(%) | dice(%) | | |
| | | | 76.18 | 71.14 | 82.67 | 81.33 | 61.21 | 76.24 | 30.29 | 59.44 |
| ✓ | | | 77.43 | 71.60 | 83.31 | 81.75 | 61.68 | 76.52 | 30.53 | 30.68 |
| ✓ | ✓ | | 81.85 | 74.59 | 85.42 | 84.84 | 64.70 | 78.25 | 38.49 | 39.17 |
| ✓ | ✓ | ✓ | **82.01** | **76.25** | **85.96** | **85.28** | **65.84** | **78.90** | 42.21 | 41.35 |

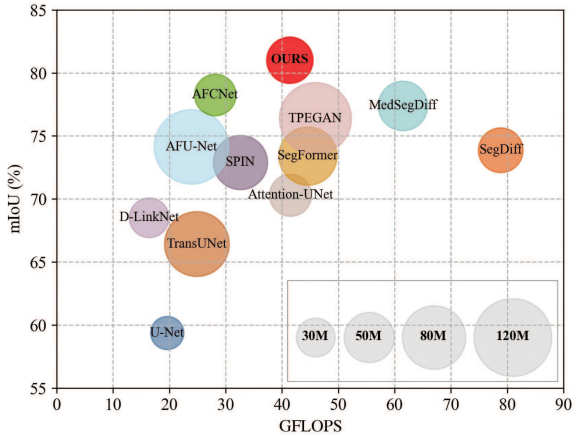


Fig. 9. Efficiency comparison of AFLDiff with competing methods on the DeepGlobe dataset. The position of each bubble represents the model's computational complexity (GLOPS) and accuracy (mIoU), and the size of each bubble indicates the model's number of parameters.

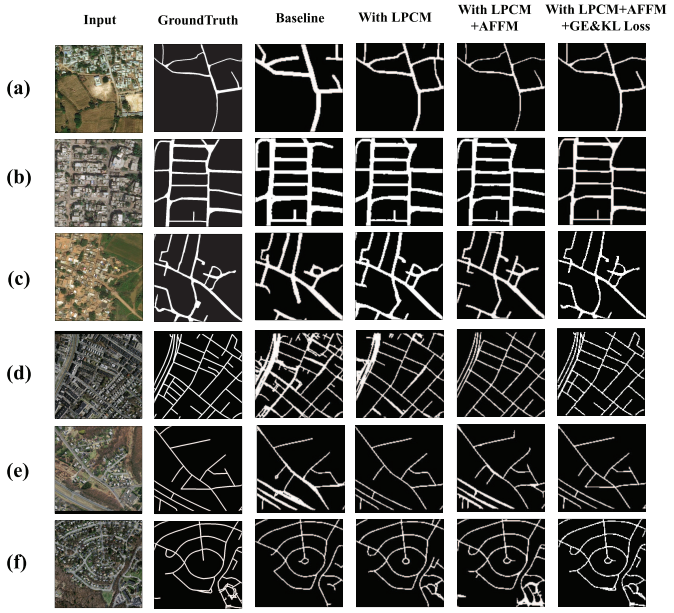


Fig. 10. Visualization results of ablation studies on modules. (a)–(c) Results from the DeepGlobe dataset. (d)–(f) Results from the Massachusetts dataset.

create various model variants. All the models are trained in the same computational environment, using the same training dataset, data preprocessing techniques, and training strategies until convergence.

The quantitative metrics of different model variants on the two datasets are shown in Table III. The introduction of LPCM improves AFLDiff's computational efficiency by 48.38% while maintaining a relatively constant number of model parameters, demonstrating that the latent space training strategy and perceptual compression for high-resolution remote sensing images significantly accelerate model computation.

AFFM makes the most significant contribution to the accuracy of the road masks generated by AFLDiff, resulting in improvements of 2.11% in dice score, 4.42% in mIoU, and 2.99% in rIoU on the DeepGlobe dataset, with 1.67% in dice score, 3.09% in mIoU, and 2.06% in rIoU on the Massachusetts dataset. This shows that adaptive frequency-aware filtering amplifies useful low-frequency and high-frequency information within noisy features while filtering out irrelevant and redundant components, thereby narrowing the gap between noisy features and pixel features. The resulting fused features significantly advance the diffusion model's ability to learn the sampling probability distribution.

In addition, the KL divergence loss defined by the Gaussian encoder provides a measurable boost to model performance, indicating that modeling noise and its resultant heterogeneity within the probability distribution enhance the diffusion model's capacity to generate higher quality predictions for road masks.

From the qualitative visualization results, illustrated in Fig. 10, it can be seen that all the three modules contribute to improving the quality of the generated road masks. Specifically, the introduction of the LPCM module significantly enhances the refinement of the generated masks, effectively recognizing scale variations in road features. This improvement is attributed to LPCM, which greatly increases the attention scores for road areas, while the pyramid structure models multiscale pixel features. The AFFM module significantly enhances the accuracy of mask generation and corrects errors caused by noise interference in the masks, demonstrating that adaptive frequency-aware filtering effectively eliminates noise disturbances and achieves accurate and robust feature fusion. In addition, the Gaussian encoders with KL divergence have unique advantages in optimizing details. For instance, in the results of row (c) and (f), the KL loss makes the curvature of complex ring-shaped roads more closely aligned with the ground-truth masks.

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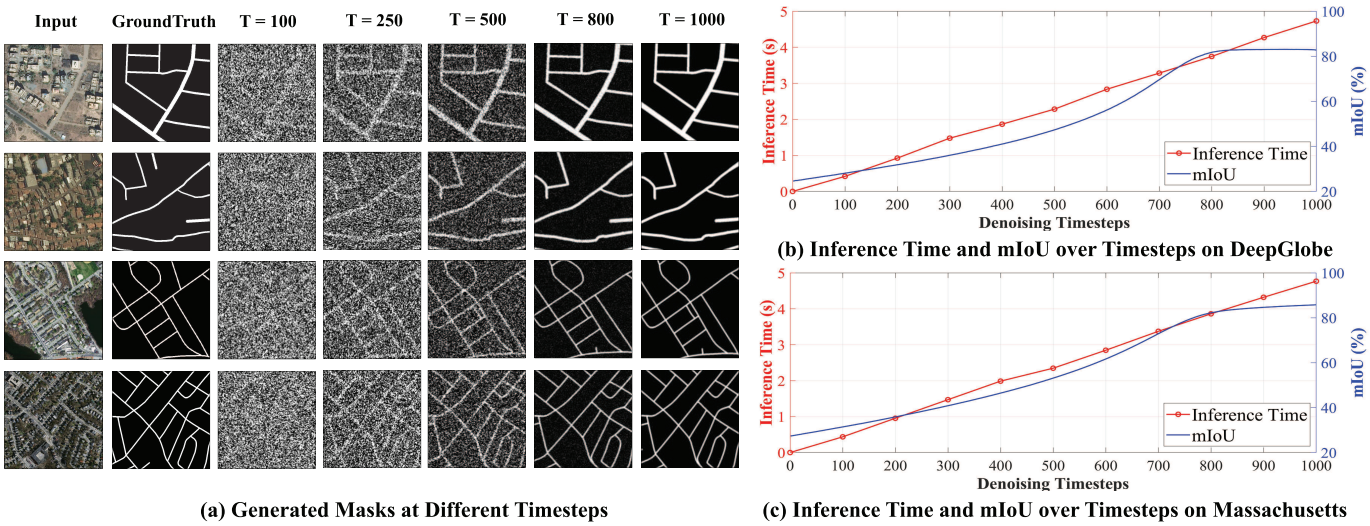


(a) Generated Masks at Different Timesteps

(b) Inference Time and mIoU over Timesteps on DeepGlobe

(c) Inference Time and mIoU over Timesteps on Massachusetts

Fig. 11. Evaluations for ablation studies on timesteps. (a) Road masks generated by AFLDiff at specified time steps. The top two rows display results from the DeepGlobe dataset, and the bottom two rows depict results from the Massachusetts dataset. (b) Variation in AFLDiff's inference time and accuracy with timesteps on the DeepGlobe dataset. (c) Variation in AFLDiff's inference time and accuracy with timesteps on the Massachusetts dataset.

TABLE IV

ABLATION STUDY RESULTS ON LOSSES USING THE DEEPGLOBE AND MASSACHUSETTS DATASETS. THE BEST RESULTS ARE HIGHLIGHTED

| MSE | CE | KL | DeepGlobe | | | Massachusetts | | |
|---|---|---|---|---|---|---|---|---|
| | | | mIoU(%) | rIoU(%) | dice(%) | mIoU(%) | rIoU(%) | dice(%) |
| ✓ | | | 74.87 | 70.85 | 83.63 | 79.72 | 58.25 | 72.69 |
| ✓ | ✓ | | 79.61 | 74.13 | 85.44 | 83.67 | 64.73 | 77.38 |
| ✓ | ✓ | ✓ | **82.01** | **76.25** | **85.96** | **85.28** | **65.84** | **78.90** |

*2) Ablation Studies on Sampling Timesteps:* For diffusion models, the number of sampling timesteps represents the iterations taken during the process of generating denoised masks, which directly affects the inference time and significantly influences prediction accuracy. It is essential to find a balance between accuracy and computational efficiency by choosing an appropriate number of timesteps. In this section, we fix the time steps at 100, 250, 500, 800, and 1000 on both the datasets, and the prediction results of AFLDiff in these timesteps are illustrated in Fig. 11.

As shown in Fig. 11(a), during the progression of denoising steps, AFLDiff gradually removes noise starting from random noise, refining the generation of denoised masks. By the 800th denoising step, the diffusion model begins to produce acceptable results, though there are still some faint noise spots. At the 1000th timestep, most of the noise in the generated mask is completely eliminated, with the road areas being accurate and rich in detail. Fig. 11(b) and (c) displays the changes in inference time and mIoU of the predicted road masks by AFLDiff on the DeepGlobe and Massachusetts datasets over time steps. From the variation in mIoU, it can be seen that AFLDiff reaches accuracy saturation between the 800th and 1000th timesteps on both the datasets. Moreover, from the intersection point of the two lines, it can be observed that AFLDiff achieves an optimal balance between accuracy and inference speed around the 800th timestep.

*3) Ablation Studies on Losses:* During the training process of AFLDiff, in addition to the mse loss defined in the original DDPM for noise, we also incorporate a CE loss for the predicted road masks and a KL divergence loss for the probability distribution of the denoised masks throughout the process. To investigate the impact of these three loss components on the accuracy of AFLDiff's road mask generation, we conduct ablation experiments on loss functions using the two datasets. The results are shown in Table IV, where mse, CE, and KL represent the above three types of loss functions, respectively.

It can be observed that among all the losses, the CE loss contributes the most to the improvement of the metrics. The semantic segmentation loss directly influences the resulting masks, encouraging them to progressively approach the expression patterns of the ground truth. On the other hand, the KL divergence loss significantly enhances the mIoU, indicating its ability to refine the scale of roads in the predicted masks, thereby greatly improving the prediction accuracy of background pixels.

*4) Ablation Studies on Downsampling Factor:* The downsampling factor $m$ determines the compression level of the original image during latent variable encoding. To identify the optimal downsampling factor that strikes a good balance between inference speed and segmentation accuracy, we designed an ablation study on downsampling factor. By setting $m = \{2, 4, 8, 16\}$, the performance of different AFLDiff variants on the DeepGlobe dataset was assessed.

The results of the ablation study on downsampling factor are shown in Table V. It can be seen that AFLDiff ($m = 2$) achieves consistently optimal accuracy. AFLDiff ($m = 4$) achieves the best tradeoff between inference speed and segmentation performance, doubling the inference speed compared with AFLDiff ($m = 2$) while maintaining relatively high mask generation quality. However, when $m$ is increased to 16, the excessive compression level during the latent space projection in VQ-VAE severely degrades the quality of road

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TABLE V
QUANTITATIVE EVALUATIONS FOR ABLATION STUDY ON DOWNSAMPLING FACTOR ON THE DEEPGLOBE DATASET

| Method | mIoU (%)↑ | rIoU (%)↑ | dice (%)↑ | Throughput (samples/s)↑ |
|---|---|---|---|---|
| AFLDiff ($m$=2) | 82.01 | 76.25 | 85.96 | 5.33 |
| AFLDiff ($m$=4) | 81.13 | 74.90 | 84.51 | 11.67 |
| AFLDiff ($m$=8) | 78.12 | 71.68 | 82.07 | 18.82 |
| AFLDiff ($m$=16) | 70.94 | 63.30 | 75.80 | 29.46 |

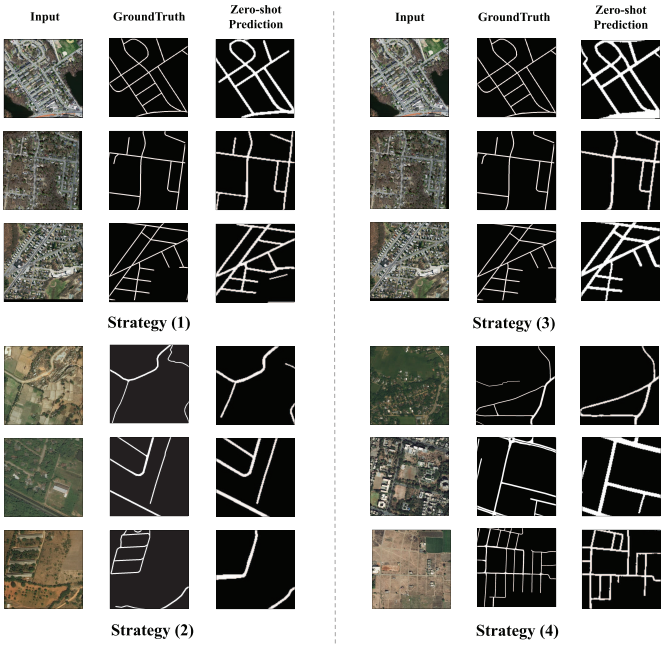


Fig. 12. Visualization results of cross-dataset validation.The left side shows the visualization results of the original DeepGlobe and Massachusetts cross-validation, while the right side presents the visualization results of the cross-validation between the downsampled DeepGlobe and Massachusetts.

images, damaging pixel features and ultimately leading to a significant drop in segmentation accuracy.

### D. Cross-Dataset Validation

Within remote sensing images, the background is complex, and the geometric shapes and scales of the roads vary significantly. Therefore, robustness is a crucial factor for the application of road extraction models to real-world sample problems. To systematically validate the generalization ability of the proposed AFLDiff, we use the following strategy: we train the model on one public dataset and then validate it on another dataset.

Notably, the spatial resolutions of the DeepGlobe and Massachusetts datasets are 0.5 and 1.2 m, respectively. To investigate the impact of aligning ground resolution on cross-dataset validation performance, we follow the approach in [69] and resize the original 1024 × 1024 images in the DeepGlobe dataset to 512 × 512, resulting in a downsampled version with a spatial resolution of 1.0 m. The resulting visualizations are shown in Fig. 12, and the quantitative results of the metrics are presented in Table VI.

TABLE VI
QUANTITATIVE EVALUATIONS FOR CROSS-DATASET VALIDATION

| Strategy | mIoU(%) | rIoU(%) | dice(%) |
|---|---|---|---|
| **(1)** Train: DeepGlobe | | | |
| Inference: Massachusetts | 68.52 | 52.77 | 61.83 |
| **(2)** Train: Massachusetts | | | |
| Inference: DeepGlobe | 65.37 | 58.41 | 65.90 |
| **(3)** Train: DeepGlobe (Downsampled) | | | |
| Inference: Massachusetts | 65.29 | 51.06 | 56.77 |
| **(4)** Train: Massachusetts | | | |
| Inference: DeepGlobe (Downsampled) | 65.71 | 58.59 | 66.36 |

The results show that when validating on a different dataset, AFLDiff's prediction accuracy is somewhat reduced due to the inherent differences in road patterns between the datasets. However, it still maintains a high overall performance. For roads with minimal visual obstructions or those that are less similar to surrounding features, AFLDiff is able to generate accurate results, while still facing some difficulty in maintaining the original scale of the roads. One potential reason accounting for this could be the differing annotation standards for road scales between the two datasets.

The experimental results on the downsampled DeepGlobe dataset indicate that the transferability of AFLDiff trained on the downsampled DeepGlobe dataset significantly decreases when applied to the Massachusetts dataset. This decline is primarily due to the substantial reduction in the training set size after downsampling via image scaling. Specifically, when the resize method is used instead of the overlapping windows cropping method, the number of images in the training set decreases by nearly eight times. On the other hand, the transferability of AFLDiff trained on the Massachusetts dataset to the downsampled DeepGlobe dataset shows a slight improvement, with 0.34% in mIoU, 0.18% in rIoU, and 0.46% in dice score. Since each test image after downsampling contains richer road network information, aligning spatial resolution differences contributes to improving cross-dataset transfer performance.

Cross-dataset validation experiments demonstrate that AFLDiff exhibits strong robustness in road extraction tasks, largely attributed to its powerful pixel distribution modeling capabilities as a generative model. These findings provide a strong basis for further exploration of transfer learning, semi-supervised, and even unsupervised training strategies to optimize diffusion models for road extraction across diverse scenarios.

## VI. CONCLUSION

In this article, we propose an innovative AFLDiff for road extraction in remote sensing images. To bridge the gap between rich semantic features and noisy features for better fusion, the AFFM uses an adaptive frequency-aware filter to remove redundant frequency components and uses an

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attention mechanism to enhance the effective components. To reduce the computational resource consumption when modeling complex features in high-resolution images with diffusion models, the LPCM uses a coupled perception compression VQ-VAE encoder to model low-dimensional latent space variables. Moreover, to improve the loss function, at randomly sampled time steps, we use Gaussian encoders to model the probability distributions of the predicted latent variables, progressively refining mask denoising by reducing distribution heterogeneity. Extensive experiments on three public datasets demonstrate that the proposed AFLDiff consistently achieves SOTA performance, and the cross-dataset validation experiments have confirmed the robustness and generalization capabilities of AFLDiff. In the future, we will further explore semi-supervised and even unsupervised training strategies to enhance AFLDiff's application in few-shot or zero-shot tasks and extend its use to real-world road extraction scenarios. In addition, further investigating loss constraint methods for latent probability distributions could serve as a promising direction to strengthen diffusion models' comprehension of complex road structures.

## REFERENCES


[1] W. Wang, N. Yang, Y. Zhang, F. Wang, T. Cao, and P. Eklund, "A review of road extraction from remote sensing images," *J. Traffic Transp. Eng.*, vol. 3, no. 3, pp. 271–282, 2016.

[2] R. Lian, W. Wang, N. Mustafa, and L. Huang, "Road extraction methods in high-resolution remote sensing images: A comprehensive review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5489–5507, 2020.

[3] L. Chen et al., "Milestones in autonomous driving and intelligent vehicles: Survey of surveys," *IEEE Trans. Intell. Vehicles*, vol. 8, no. 2, pp. 1046–1056, Feb. 2023.

[4] L. Ding and L. Bruzzone, "DiResNet: Direction-aware residual network for road extraction in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, pp. 10243–10254, 2021.

[5] Z. Xu et al., "RNGDet: Road network graph detection by transformer in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4707612, doi: 10.1109/TGRS.2022.3186993.

[6] L. Dai, G. Zhang, and R. Zhang, "RADANet: Road augmented deformable attention network for road extraction from complex high-resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023Art. no. 5602213, doi: 10.1109/TGRS.2023.3237561.

[7] R. Liu, F. Li, W. Jiang, C. Song, Q. Chen, and Z. Li, "Generating pixel enhancement for road extraction in high-resolution aerial images," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 10, pp. 6313–6325, Oct. 2024.

[8] Z. Lai et al., "Denoising diffusion semantic segmentation with mask prior modeling," 2023, *arXiv:2306.01721*.

[9] S. Hong, G. Lee, W. Jang, and S. Kim, "Improving sample quality of diffusion models using self-attention guidance," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 7462–7471.

[10] W. Peebles and S. Xie, "Scalable diffusion models with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4195–4205.

[11] C. Vignac et al., "DiGress: Discrete denoising diffusion for graph generation," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 16506–16527.

[12] L. Zhang and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 3836–3847.

[13] T. Chen, C. Wang, and H. Shan, "BerDiff: Conditional Bernoulli diffusion model for medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*, 2023, pp. 491–501.

[14] H. Wang, J. Cao, R. M. Anwer, J. Xie, F. S. Khan, and Y. Pang, "DFormer: Diffusion-guided transformer for universal image segmentation," 2023, *arXiv:2306.03437*.

[15] J. Wu, H. Fang, Y. Zhang, Y. Yang, and Y. Xu, "MedSegDiff: Medical image segmentation with diffusion probabilistic model," in *Proc. Med. Imag. Deep Learn.*, 2022, pp. 1623–1639.

[16] X. Gu, M. Zhang, J. Lyu, and Q. Ge, "Generating urban road networks with conditional diffusion models," *ISPRS Int. J. Geo-Inf.*, vol. 13, no. 6, p. 203, Jun. 2024.

[17] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

[18] Y. Takagi and S. Nishimoto, "High-resolution image reconstruction with latent diffusion models from human brain activity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14453–14463.

[19] I. N. Sneddon, *Fourier Transforms*. Mineola, NY, USA: Dover Publications, 1995.

[20] J. Liu, Z. Yuan, Z. Pan, Y. Fu, L. Liu, and B. Lu, "Diffusion model with detail complement for super-resolution of remote sensing," *Remote Sens.*, vol. 14, no. 19, p. 4834, Sep. 2022.

[21] D. Baranchuk, I. Rubachev, A. Voynov, V. Khrulkov, and A. Babenko, "Label-efficient semantic segmentation with diffusion models," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 15633–15647.

[22] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, 2020, pp. 6840–6851.

[23] G. J. Chowdary and Z. Yin, "Diffusion transformer U-Net for medical image segmentation," in *Proc. Int. Conf. Med. image Comput. computer-assisted Intervent*, 2023, pp. 622–631.

[24] A. Van Den Oord, O. Vinyals, and K. Kavukcuoglu, "Neural discrete representation learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 6306–6315.

[25] H. Vu Quoc, T. Tran Le Phuong, M. Trinh Xuan, and S. Dinh Viet, "LSegDiff: A latent diffusion model for medical image segmentation," in *Proc. 12th Int. Symp. Inf. Commun. Technol.*, Dec. 2023, pp. 456–462.

[26] W. V. Gansbeke and B. D. Brabandere, "A simple latent diffusion approach for panoptic segmentation and mask inpainting," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 78–97.

[27] B. Dong, P. Wang, and F. Wang, "Head-free lightweight semantic segmentation with linear transformer," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 1, pp. 516–524.

[28] F. Li, R. Feng, W. Han, and L. Wang, "High-resolution remote sensing image scene classification via key filter bank based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8077–8092, Nov. 2020.

[29] Y. Zhang, W. Liu, Z. Chen, J. Wang, and K. Li, "On the properties of Kullback–Leibler divergence between multivariate Gaussian distributions," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 8521–8534.

[30] Y. Zang, C. Wang, Y. Yu, L. Luo, K. Yang, and J. Li, "Joint enhancing filtering for road network extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1511–1525, Mar. 2017.

[31] C. Poullis and S. You, "Delineation and geometric modeling of road networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 65, no. 2, pp. 165–181, Mar. 2010.

[32] A. A. Matkan, M. Hajeb, and S. Sadeghian, "Road extraction from LiDAR data using support vector machine classification," *Photogramm. Eng. Remote Sens.*, vol. 80, no. 5, pp. 409–422,May 2014.

[33] L. Xiao, B. Dai, D. Liu, D. Zhao, and T. Wu, "Monocular road detection using structured random forest," *Int. J. Adv. Robot. Syst.*, vol. 13, no. 3, p. 101, May 2016.

[34] R. Maurya, P. R. Gupta, and A. Shankar Shukla, "Road extraction using K-means clustering and morphological operations," in *Proc. Int. Conf. Image Inf. Process.*, Nov. 2011, pp. 1–6.

[35] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[36] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[37] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[38] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[39] L. Zhou, C. Zhang, and M. Wu, "D-LinkNet: LinkNet with pretrained encoder and dilated convolution for high resolution satellite imagery road extraction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 182–186.


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[40] A. Chaurasia and E. Culurciello, "LinkNet: Exploiting encoder representations for efficient semantic segmentation," in *Proc. IEEE Vis. Commun. Image Process. (VCIP)*, Dec. 2017, pp. 1–4.
[41] J. Mei, R.-J. Li, W. Gao, and M.-M. Cheng, "CoANet: Connectivity attention network for road extraction from satellite imagery," *IEEE Trans. Image Process.*, vol. 30, pp. 8540–8552, 2021.
[42] L. Luo, J.-X. Wang, S.-B. Chen, J. Tang, and B. Luo, "BDTNet: Road extraction by bi-direction transformer from remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[43] X. Liu et al., "RoadFormer: Road extraction using a Swin transformer combined with a spatial and channel separable convolution," *Remote Sens.*, vol. 15, no. 4, p. 1049, Feb. 2023.
[44] C. Ge, Y. Nie, F. Kong, and X. Xu, "Improving road extraction for autonomous driving using Swin transformer unet," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2022, pp. 1216–1221.
[45] W. G. C. Bandara, J. M. J. Valanarasu, and V. M. Patel, "SPIN road mapper: Extracting roads from aerial images via spatial and interaction space graph reasoning for autonomous driving," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 343–350.
[46] F. Bastani et al., "RoadTracer: Automatic extraction of road networks from aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4720–4728.
[47] Y.-Q. Tan, S.-H. Gao, X.-Y. Li, M.-M. Cheng, and B. Ren, "VecRoad: Point-based iterative graph exploration for road graphs extraction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8907–8915.
[48] T. Amit, T. Shaharbany, E. Nachmani, and L. Wolf, "SegDiff: Image segmentation with diffusion probabilistic models," 2021, *arXiv:2112.00390*.
[49] Z. Gu, H. Chen, and Z. Xu, "Diffusioninst: Diffusion model for instance segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 2730–2734.
[50] A. Rahman, J. M. J. Valanarasu, I. Hacihaliloglu, and V. M. Patel, "Ambiguous medical image segmentation using diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11536–11546.
[51] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[52] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.
[53] Y. Zhong, B. Li, L. Tang, S. Kuang, S. Wu, and S. Ding, "Detecting camouflaged object in frequency domain," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4504–4513.
[54] J. Wu, W. Ji, H. Fu, M. Xu, Y. Jin, and Y. Xu, "MedSegDiff-v2: Diffusion-based medical image segmentation with transformer," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 6, pp. 6030–6038.
[55] K. Wu, H. Peng, M. Chen, J. Fu, and H. Chao, "Rethinking and improving relative position encoding for vision transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10033–10041.
[56] J. Li, H. Xie, J. Li, Z. Wang, and Y. Zhang, "Frequency-aware discriminative feature learning supervised by single-center loss for face forgery detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6458–6467.
[57] I. Demir et al., "DeepGlobe 2018: A challenge to parse the Earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 172–181.
[58] V. Mnih, "Machine learning for aerial image labeling," Ph.D. thesis, Dept. Comput. Center Toronto, Univ. Toronto, Toronto, ON, Canada, 2013.
[59] Q. Zhu et al., "A global context-aware and batch-independent network for road extraction from VHR satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 353–365, May 2021.
[60] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.
[61] Z. Zhu, Y. Yan, R. Xu, Y. Zi, and J. Wang, "Attention-UNet: A deep learning approach for fast and accurate segmentation in medical imaging," *J. Comput. Sci. Softw. Appl.*, vol. 2, no. 4, pp. 24–31, 2022.
[62] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Álvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 12077–12090.
[63] J. Yang and H. Liu, "Modulation learning on Fourier-domain for road extraction from remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[64] H. Liu, C. Wang, J. Zhao, S. Chen, and H. Kong, "Adaptive Fourier convolution network for road segmentation in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024Art. no. 5617214, doi: 10.1109/TGRS.2024.3384059.
[65] B. Zhong et al., "FERDNet: High-resolution remote sensing road extraction network based on feature enhancement of road directionality," *Remote Sens.*, vol. 17, no. 3, p. 376, Jan. 2025. [Online]. Available: https://www.mdpi.com/2072-4292/17/3/376
[66] J. Yang, T. Liu, L. Zhang, and Y. Zhang, "Effective road segmentation with selective state-space model and frequency feature compensation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025Art. no. 5603813, doi: 10.1109/TGRS.2024.3521483.
[67] X. Bai and L. Melas-Kyriazi, "Fixed point diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 9430–9440.
[68] T. Guo, Y. Gao, F. Luo, L. Zhang, B. Du, and X. Gao, "DENet: Direction and edge co-awareness network for road extraction from high-resolution remote sensing imagery," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 7, pp. 1–14, Jul. 2025.
[69] R. Huang, M. Ma, and W. Huang, "Domain adaptation for multilabel remote sensing image annotation with contrastive pseudo-label generation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 20344–20354, 2024.

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